

Hierarchical Product Catalog

20 products with category, subcategory, and variant relationships

Designed as a text-only corpus for hierarchical RAG learning and validation

Catalog hierarchy index

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- HCL-101 - AeroTherm X1 (children: HCL-101-S, HCL-101-P)
- HCL-102 - BreezeSense V2 (children: HCL-102-B, HCL-102-A)

Home Systems

Security and Access

- HSC-201 - Sentinel Lock S4 (children: HSC-201-K, HSC-201-F)
- HSC-202 - WatchTower Cam 4K (children: HSC-202-I, HSC-202-O)

Home Systems

Water Protection

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Office Technology

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Office Technology

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Mobility

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Home Appliances

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Home Appliances

Wellness

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- APW-1002 - SleepLight Rhythm (children: no variants)

RAG hint: retain category, subcategory, product ID, and variant ID as separate metadata fields.

AeroTherm X1

Product ID: HCL-101 | Manufacturer: HelioWorks Climate | Starting price: \$249

Home Systems > Climate Control > AeroTherm X1

Purpose and scope

A room-scale smart heat-pump controller for quiet, schedule-based heating and cooling in apartments and small offices.

Specifications

Coverage: Up to 45 m2 | Power: 24V control circuit | Connectivity: Wi-Fi 6 / Matter | Warranty: 3 years

Variants

- HCL-101-S - Standard - \$249: Single-zone controller with wall sensor.
- HCL-101-P - Plus - \$319: Adds remote occupancy sensor and humidity control.

Quick start

1. Switch off HVAC control power.
2. Connect labelled terminals and restore power.
3. Run the compatibility and sensor test.

How to use

- Choose a schedule or set a temporary target.
- Review runtime trends before changing limits.

Error codes and resolution

CLM-E101: Sensor unavailable

Reseat the wall sensor and run Sensor Test.

CLM-E214: Safety delay active

Wait five minutes; do not switch modes repeatedly.

CLM-E330: Zone conflict

Check schedule overlap and the heating/cooling deadband.

BreezeSense V2

Product ID: HCL-102 | Manufacturer: HelioWorks Climate | Starting price: \$189

Home Systems > Climate Control > BreezeSense V2

Purpose and scope

A connected ventilation controller that balances fresh-air intake, fan speed, and humidity targets for a single indoor zone.

Specifications

Airflow: 90-220 m3/h | Power: 100-240V AC | Connectivity: Wi-Fi 5 | Warranty: 2 years

Variants

- HCL-102-B - Base - \$189: Manual fan curves and app alerts.
- HCL-102-A - Auto - \$239: Adds CO2-driven fan control and weekly reports.

Quick start

1. Mount away from direct exhaust flow.
2. Connect fan and power leads to labelled terminals.
3. Set preferred CO2 and humidity thresholds.

How to use

- Use Auto mode for normal operation.
- Select Boost mode for short high-occupancy periods.

Error codes and resolution

CLM-E118: Airflow sensor blocked

Clean the intake grille and restart calibration.

CLM-E227: Fan response timeout

Check breaker, wiring, and fan service switch.

CLM-E344: Humidity reading unstable

Move the sensor away from steam or open windows.

Sentinel Lock S4

Product ID: HSC-201 | Manufacturer: Aegis Entry Systems | Starting price: \$219

Home Systems > Security and Access > Sentinel Lock S4

Purpose and scope

A keypad smart deadbolt for residential and small-office doors, with role-based entry codes and an audit trail.

Specifications

Codes: Up to 100 | Power: 4 AA batteries | Door thickness: 35-55 mm | Warranty: 3 years

Variants

- HSC-201-K - Keypad - \$219: Keypad, mechanical key, and mobile unlock.
- HSC-201-F - Keypad + Fob - \$269: Adds encrypted fob access for staff or guests.

Quick start

1. Install with the door open and bolt fully aligned.
2. Create owner and backup codes.
3. Test lock, unlock, and auto-lock before closing the door.

How to use

- Use temporary codes for visitors.
- Review access history after a security event.

Error codes and resolution

SEC-E101: Bolt obstructed

Open the door and correct strike alignment.

SEC-E205: Code rejected

Confirm that the code is active and not expired.

SEC-E318: Battery critical

Use the mechanical key and replace all batteries.

WatchTower Cam 4K

Product ID: HSC-202 | Manufacturer: Aegis Entry Systems | Starting price: \$329

[Home Systems](#) > [Security and Access](#) > [WatchTower Cam 4K](#)

Purpose and scope

A 4K entry camera with person detection, local recording, privacy masks, and administrator-only clip export.

Specifications

Video: 4K HDR | Night vision: Up to 10 m | Storage: Local microSD / recorder | Weather rating: IP65

Variants

- HSC-202-I - Indoor - \$329: Desk or wall mount with privacy shutter.
- HSC-202-O - Outdoor - \$369: Weather-sealed body and PoE option.

Quick start

1. Mount with a clear view and stable network signal.
2. Apply updates before enabling recording.
3. Draw privacy masks before defining motion zones.

How to use

- Review Events for detected activity.
- Export only required clips and keep access roles current.

Error codes and resolution

SEC-E142: Stream unavailable

Check power, network link, and recorder status.

SEC-E253: Storage nearly full

Archive required clips or revise retention policy.

SEC-E367: Privacy mask conflict

Separate the motion zone from masked pixels.

AquaGuard Flow

Product ID: HWT-301 | Manufacturer: ClearFlow Technologies | Starting price: \$299

Home Systems > Water Protection > AquaGuard Flow

Purpose and scope

A whole-home flow monitor that identifies sustained unusual water use and can notify designated household members.

Specifications

Pipe sizes: 15-32 mm | Detection: Continuous flow analysis | Power: USB-C / adapter | Warranty: 2 years

Variants

- HWT-301-M - Monitor - \$299: Monitoring and alerting only.
- HWT-301-S - Shutoff Kit - \$469: Adds installer-fitted automatic shutoff valve.

Quick start

1. Fit the sensor around the labelled supply pipe.
2. Run a baseline flow calibration with all fixtures closed.
3. Name the monitored property and alert recipients.

How to use

- Review Flow History before classifying a use as unusual.
- Use Maintenance Mode during planned plumbing work.

Error codes and resolution

WTR-E101: Active flow alert

Inspect fixtures and appliances for continuous water use.

WTR-E238: Calibration incomplete

Repeat calibration when no water is running.

WTR-E355: Valve response failed

Use manual override and contact a qualified plumber.

LeakNode Mini

Product ID: HWT-302 | Manufacturer: ClearFlow Technologies | Starting price: \$39

Home Systems > Water Protection > LeakNode Mini

Purpose and scope

A compact battery leak sensor for cabinets, water heaters, washing machines, and other localized risk points.

Specifications

Battery: CR2450 coin cell | Alert: Local alarm + app | Temperature: 0-50 C | Warranty: 1 year

Variants

- This product has no child variant nodes. Use the product ID as the most specific catalog node.

Quick start

1. Place contacts flat on a dry surface.
2. Label the exact room and appliance in the app.
3. Test with a few drops of clean water, then dry contacts.

How to use

- Treat every wet-sensor alert as urgent.
- Replace the battery when the app prompts you.

Error codes and resolution

WTR-E109: Sensor wet

Resolve the leak, dry contacts, then reset.

WTR-E219: Battery low

Replace with the specified coin cell.

WTR-E341: Connection weak

Move the hub or add a compatible repeater.

ViewPanel 55

Product ID: OFC-401 | Manufacturer: Meridian Workplace | Starting price: \$1,499

Office Technology > Collaboration > ViewPanel 55

Purpose and scope

A 55-inch collaboration display for meeting rooms, with wireless presentation, digital whiteboard, and managed guest access.

Specifications

Display: 55-inch 4K | Inputs: USB-C, HDMI, Wi-Fi | Touch: 20-point | Warranty: 3 years

Variants

- OFC-401-W - Wall - \$1,499: Wall-mount package for fixed rooms.
- OFC-401-M - Mobile - \$1,699: Adds rolling stand and cable tray.

Quick start

1. Mount securely or lock the mobile stand.
2. Connect network and register the room name.
3. Set guest-presentation and whiteboard retention policies.

How to use

- Start a session from the Room panel.
- Clear the whiteboard after each confidential meeting.

Error codes and resolution

COL-E101: No display input

Check source cable and selected input.

COL-E212: Wireless pairing failed

Confirm guest mode and network isolation settings.

COL-E329: Touch calibration required

Run calibration from Admin Settings.

VoiceDock Pro

Product ID: OFC-402 | Manufacturer: Meridian Workplace | Starting price: \$279

Office Technology > Collaboration > VoiceDock Pro

Purpose and scope

A USB and Bluetooth speakerphone for small meeting spaces, with beamforming microphones and call-quality diagnostics.

Specifications

Microphones: 6 beamforming | Range: Up to 5 m | Battery: 18 hours | Warranty: 2 years

Variants

- This product has no child variant nodes. Use the product ID as the most specific catalog node.

Quick start

1. Place centrally on a stable table.
2. Connect by USB for managed-room use or pair by Bluetooth.
3. Run the microphone test before the first meeting.

How to use

- Press Mute to disable microphones locally.
- Use Diagnostics after repeated call-quality complaints.

Error codes and resolution

COL-E115: Microphone muted

Check physical Mute button and meeting-app mute state.

COL-E245: Bluetooth pairing rejected

Remove old pairing records and pair again.

COL-E372: Echo detected

Move device away from display speakers and reduce volume.

PowerCore Desk 8

Product ID: OFP-501 | Manufacturer: Meridian Workplace | Starting price: \$149

Office Technology > Power and Connectivity > PowerCore Desk 8

Purpose and scope

A managed desk power unit with eight AC outlets, USB-C charging, individual outlet names, and energy-use reporting.

Specifications

Outlets: 8 AC + 2 USB-C | Max load: 1800 W | Monitoring: Per outlet | Warranty: 2 years

Variants

- OFP-501-D - Desk - \$149: Clamp mount and 1.8 m cord.
- OFP-501-R - Rack - \$179: 19-inch rack ears and rear outlet layout.

Quick start

1. Connect directly to a grounded wall outlet.
2. Label each outlet by connected equipment.
3. Set schedules only for devices that can safely lose power.

How to use

- Review standby usage weekly.
- Use remote restart only for approved equipment.

Error codes and resolution

PWR-E101: Overload protection active

Remove high-wattage devices and press Reset.

PWR-E224: Outlet unavailable

Enable the named outlet and inspect device power.

PWR-E339: Surge protection expired

Replace the unit; protection is no longer reliable.

LinkMesh 6E

Product ID: OFP-502 | Manufacturer: Meridian Workplace | Starting price: \$229

[Office Technology](#) > [Power and Connectivity](#) > [LinkMesh 6E](#)

Purpose and scope

A Wi-Fi 6E mesh node for managed offices, providing wired uplink, guest network support, and device-health reporting.

Specifications

Wi-Fi: Tri-band Wi-Fi 6E | Ethernet: 2.5 GbE | Coverage: Up to 180 m2 | Warranty: 3 years

Variants

- OFP-502-N - Node - \$229: Single mesh expansion node.
- OFP-502-2P - 2-Pack - \$419: Two nodes for multi-room coverage.

Quick start

1. Connect the first node to the router by Ethernet.
2. Place additional nodes within reliable signal range.
3. Create separate staff and guest network profiles.

How to use

- Check mesh health before moving a node.
- Run a network test after firmware updates.

Error codes and resolution

NET-E101: Uplink disconnected

Check Ethernet connection and router port.

NET-E216: Node placement poor

Move the node closer to a powered mesh node.

NET-E348: Guest network unavailable

Review VLAN and guest-access settings.

VoltRide City

Product ID: MOB-601 | Manufacturer: Northline Mobility | Starting price: \$1,299

Mobility > E-Bikes > VoltRide City

Purpose and scope

A commuter e-bike with pedal assist, integrated lights, removable battery, and service reminders for everyday city travel.

Specifications

Motor: 250 W hub motor | Range: Up to 65 km | Battery: 36V 10Ah | Weight: 22 kg

Variants

- MOB-601-C - City - \$1,299: Upright handlebar and rear rack.
- MOB-601-T - Touring - \$1,549: Adds suspension fork and extended battery.

Quick start

1. Charge the battery indoors using the supplied charger.
2. Adjust saddle height and test brake levers.
3. Pair the bike app and record the frame number.

How to use

- Start in low assist until familiar with handling.
- Lock battery and frame when parked.

Error codes and resolution

BIK-E101: Motor assist unavailable

Check battery charge and seat the battery fully.

BIK-E235: Brake sensor active

Release brake levers and inspect for damage.

BIK-E362: Service interval reached

Book a service inspection before intensive use.

CargoGlide 2

Product ID: MOB-602 | Manufacturer: Northline Mobility | Starting price: \$2,099

Mobility > E-Bikes > CargoGlide 2

Purpose and scope

A long-tail cargo e-bike for shopping, child-seat compatible loads, and short local deliveries, with reinforced frame and load-aware assist.

Specifications

Motor: 500 W mid-drive | Load rating: 180 kg total | Range: Up to 80 km | Brakes: Hydraulic disc

Variants

- This product has no child variant nodes. Use the product ID as the most specific catalog node.

Quick start

1. Have a dealer check frame, brakes, and cargo rail fittings.
2. Charge battery fully and calibrate load-aware assist.
3. Install approved cargo accessories only.

How to use

- Secure load low and evenly on the frame.
- Use lower assist when starting with a heavy load.

Error codes and resolution

BIK-E125: Load sensor unavailable

Remove load, restart, and schedule dealer inspection.

BIK-E248: Battery temperature high

Stop riding, shade the battery, and allow it to cool.

BIK-E381: Brake pad worn

Replace pads before further cargo use.

PackTrack One

Product ID: MOT-701 | Manufacturer: Northline Mobility | Starting price: \$69

Mobility > Travel Accessories > PackTrack One

Purpose and scope

A compact Bluetooth travel tag with last-seen location, separation alerts, and replaceable battery for luggage and equipment cases.

Specifications

Network: Bluetooth LE | Battery: CR2032 | Water rating: IP67 | Warranty: 1 year

Variants

- MOT-701-T - Tag - \$69: Single tracking tag.
- MOT-701-4 - Four Pack - \$229: Four tags with labeled holders.

Quick start

1. Remove battery tab and pair from the travel app.
2. Name the item and set a separation alert distance.
3. Test the sound function before travel.

How to use

- Use Last Seen as an approximate location, not live GPS.
- Replace battery when low-battery notification appears.

Error codes and resolution

TRV-E101: Tag not found

Enable Bluetooth and move within nearby range.

TRV-E208: Sound unavailable

Replace battery and confirm the tag is paired.

TRV-E326: Location stale

Check phone network and wait for a participating device update.

RouteCharge 65

Product ID: MOT-702 | Manufacturer: Northline Mobility | Starting price: \$79

Mobility > Travel Accessories > RouteCharge 65

Purpose and scope

A 65 W USB-C travel charger with interchangeable regional plugs and intelligent power allocation for laptops, phones, and accessories.

Specifications

Output: 65 W USB-C PD | Ports: 2 USB-C + 1 USB-A | Input: 100-240V | Warranty: 2 years

Variants

- This product has no child variant nodes. Use the product ID as the most specific catalog node.

Quick start

1. Fit the correct regional plug until it clicks.
2. Connect highest-power device to USB-C port 1.
3. Check that the wall outlet is undamaged and compatible.

How to use

- Disconnect before changing plug heads.
- Use certified cables appropriate for device power requirements.

Error codes and resolution

TRV-E120: Power negotiation failed

Reconnect cable and use a certified USB-C PD cable.

TRV-E236: Temperature protection active

Unplug, let charger cool, and reduce connected load.

TRV-E349: Plug head not locked

Remove and refit the plug head until fully engaged.

BrewCraft Duo

Product ID: APK-801 | Manufacturer: Harborline Appliances | Starting price: \$189

Home Appliances > Kitchen > BrewCraft Duo

Purpose and scope

A dual-mode coffee maker with drip and single-serve functions, programmable start, strength control, and cleaning reminders.

Specifications

Capacity: 1.5 L carafe | Modes: Drip / single serve | Power: 1450 W | Warranty: 2 years

Variants

- APK-801-G - Glass Carafe - \$189: Heated glass carafe version.
- APK-801-T - Thermal Carafe - \$219: Insulated thermal carafe version.

Quick start

1. Rinse removable parts and run one water-only cycle.
2. Set local time and preferred strength.
3. For scheduled brew, prepare water and grounds before bedtime.

How to use

- Use single-serve mode with the correct adapter.
- Run cleaning cycle when the maintenance alert appears.

Error codes and resolution

KIT-E101: Water level too low

Fill reservoir above the minimum mark.

KIT-E223: Carafe not detected

Center carafe and clean the warming plate.

KIT-E347: Descale required

Run the guided descale cycle before brewing again.

FreshVault 12

Product ID: APK-802 | Manufacturer: Harborline Appliances | Starting price: \$599

Home Appliances > Kitchen > FreshVault 12

Purpose and scope

A compact 12-bottle preservation cabinet with adjustable temperature zones, humidity display, and door-open notifications.

Specifications

Capacity: 12 bottles | Zones: Dual temperature | Range: 5-18 C | Warranty: 3 years

Variants

- This product has no child variant nodes. Use the product ID as the most specific catalog node.

Quick start

1. Place on a level surface with rear ventilation clearance.
2. Allow two hours of stabilization before loading.
3. Set upper and lower zone targets, then enable door alerts.

How to use

- Arrange bottles without blocking internal airflow.
- Review temperature history after a power interruption.

Error codes and resolution

KIT-E134: Door open too long

Close door fully and inspect gasket alignment.

KIT-E251: Zone sensor fault

Restart appliance and request service if fault returns.

KIT-E369: Compressor protection delay

Wait for delay to finish before changing target again.

CleanSweep R3

Product ID: APC-901 | Manufacturer: Harborline Appliances | Starting price: \$429

Home Appliances > Cleaning > CleanSweep R3

Purpose and scope

A robot vacuum for hard floors and low-pile carpets, with room mapping, no-go zones, scheduled runs, and washable filter.

Specifications

Runtime: 150 minutes | Bin: 0.45 L | Maps: Up to 5 floors | Warranty: 2 years

Variants

- APC-901-V - Vacuum - \$429: Vacuum-only unit.
- APC-901-M - Vacuum + Mop - \$499: Adds detachable controlled-flow mop module.

Quick start

1. Place dock against a clear wall and charge fully.
2. Map each floor with cords and loose objects removed.
3. Set no-go zones for stairs, cables, and pet areas.

How to use

- Choose rooms or a whole-floor task.
- Empty bin after major runs and dry mop pad after use.

Error codes and resolution

CLN-E101: Robot stuck

Clear obstacle and resume from open floor.

CLN-E218: Dock not found

Check dock power and approach path.

CLN-E356: Brush jam

Remove hair or thread from brush and end caps.

SteamMate S1

Product ID: APC-902 | Manufacturer: Harborline Appliances | Starting price: \$159

Home Appliances > Cleaning > SteamMate S1

Purpose and scope

A compact garment steamer with quick heat-up, removable water tank, and auto-off protection for household clothing care.

Specifications

Heat-up: 35 seconds | Tank: 220 ml | Steam modes: 2 | Warranty: 1 year

Variants

- This product has no child variant nodes. Use the product ID as the most specific catalog node.

Quick start

1. Fill tank with clean water and lock it in place.
2. Power on while upright and wait for ready indicator.
3. Test steam away from garments first.

How to use

- Keep fabric taut and move slowly from top to bottom.
- Empty tank after use and allow unit to cool upright.

Error codes and resolution

CLN-E125: Water tank missing

Refit tank until it clicks.

CLN-E242: Mineral buildup

Use the approved cleaning process; do not use vinegar.

CLN-E371: Overheat protection

Power off, let cool, and inspect air vents.

AirPure Cube

Product ID: APW-1001 | Manufacturer: Harborline Appliances | Starting price: \$279

Home Appliances > Wellness > AirPure Cube

Purpose and scope

A HEPA air purifier for bedrooms and workspaces, with particle sensing, automatic fan speed, and filter-life reporting.

Specifications

Coverage: Up to 55 m2 | Filter: HEPA H13 + carbon | Noise: 24-52 dB | Warranty: 2 years

Variants

- APW-1001-H - Home - \$279: Standard filter and display.
- APW-1001-A - Allergy - \$319: Adds enhanced carbon filter and pollen mode.

Quick start

1. Remove filter packaging before first use.
2. Place 30 cm from walls or furniture.
3. Select Auto mode and allow 20 minutes for baseline reading.

How to use

- Use Sleep mode overnight for lower noise.
- Replace filter only when prompted or airflow falls noticeably.

Error codes and resolution

AIR-E101: Filter not detected

Remove packaging and reseal filter panel.

AIR-E231: Particle sensor dirty

Clean sensor inlet with dry soft brush.

AIR-E358: Airflow reduced

Inspect filter condition and intake clearance.

SleepLight Rhythm

Product ID: APW-1002 | Manufacturer: Harborline Appliances | Starting price: \$119

Home Appliances > Wellness > SleepLight Rhythm

Purpose and scope

A bedside circadian light with timed warm-to-cool transitions, wake-up routines, and a simple local control dial.

Specifications

Brightness: 1-900 lumens | Color range: 1800-6500 K | Power: USB-C | Warranty: 2 years

Variants

- This product has no child variant nodes. Use the product ID as the most specific catalog node.

Quick start

1. Place on a stable bedside surface.
2. Set local time and preferred wake window.
3. Create a gentle morning routine before enabling it.

How to use

- Turn the dial for brightness and press to change mode.
- Use wind-down routine in the hour before sleep.

Error codes and resolution

AIR-E117: Clock not set

Set time in the companion app or local menu.

AIR-E247: Routine skipped

Check schedule, time zone, and power connection.

AIR-E374: LED temperature warning

Reduce brightness, allow cooling, and check ventilation.